

BENCH-SCALE PHASE-CHANGE AND INTUMESCENT MECHANISM TEST

PRE-REGISTERED TEST REPORT

Report number: CIP-BTSB-BT-001 Revision: A — pre-registration Issued: 25 August 2026 Author: S. Mohammed Idreesh, CIP — Compute Infrastructure Platform, Chennai

DOCUMENT STATUS

THIS DOCUMENT WAS COMPLETED AND DATED BEFORE ANY DATA WERE COLLECTED.

Sections 1-9 state the apparatus, method, quantitative predictions, pre-specified analysis and falsification criteria. Sections 10-13 are blank at issue and are completed after testing.

The purpose of issuing this document in advance is to fix the predictions and the analysis method before the results are known, so that neither can be adjusted afterwards to fit the outcome.

Revision	Date	Status	Contains results?
A	25 Aug 2026	Pre-registration	No
B	___ Aug 2026	Final	Yes

Apparatus verification completed before issue. Gate records G1 (software chain), G2 (sensing) and G3 (heater circuit) were completed and issued before this document was dated, and no heat had been applied to the test fixture at the time of issue. Those records are listed in Section 4.5 and are published alongside this report.

[PRE-GEN-0 MECHANISM DEMONSTRATION — NOT A PRODUCT VALIDATION]

1. PURPOSE

To determine, by physical measurement, whether a phase-change material layer produces a measurable delay in heat transfer across a fixed gap, and whether that delay can be attributed to latent heat rather than to thermal resistance alone.

A second, independent test measures the activation onset temperature of an expandable graphite intumescent.

Both mechanisms are constituent layers of BTSB, a five-layer passive inter-cell barrier for lithium-ion battery packs.

2. WHAT THIS TEST DOES NOT DO

Stated first, so that nothing later in the document can be read as claiming more than it does.

- **It does not test BTSB.** It tests two of BTSB’s constituent mechanisms in isolation, using proxy materials, in a fixture that is not a battery pack.
- **It does not use the specified PCM.** The specified material is RT64HC (Rubitherm). The proxy used here is fully refined paraffin wax, chosen for availability and for a melting range near the specified onset.
- **It involves no cells.** No lithium-ion cell is present. No thermal runaway is initiated. Heat is supplied by a resistive cartridge heater under approximately constant power.
- **It is not at product scale.** The bench charge holds approximately **3.3x** the latent energy of a production BTSB pad (4,600 J against 1,387 J).
- **It does not measure propagation delay,** flame propagation, gas adsorption or dielectric performance. Those require GB 38031-2025 nail penetration testing on instrumented cells at a NABL-accredited facility, planned as Gen-0.
- **It is not independently witnessed** and was not performed in an accredited laboratory.

3. BACKGROUND — THE MECHANISM UNDER TEST

A phase-change material absorbs energy at approximately constant temperature while melting. Placed between a heat source and a protected body, it acts as a temporary thermal buffer: the protected body’s temperature rise stalls for as long as unmelted material remains.

Two distinct effects arise when any layer is inserted into a heat path:

- 1. **Thermal resistance.** Any material of finite conductivity impedes heat flow. This is present whether or not the material changes phase.
- 2. **Latent heat.** Energy absorbed during melting, at constant temperature, additional to the resistance effect.

Both produce a delay. Only the second is the phase-change mechanism. A test that compares a PCM against an empty gap cannot distinguish them, and would overstate the phase-change contribution.

The method in Section 5 separates them by using a third condition — a liquid of similar composition and conductivity that does not change phase in the temperature range of interest.

4. APPARATUS

4.1 Fixture

Item	Specification
Plates	2 × 173 × 71 × 3 mm, 5052-H32 aluminium, plain mill finish
Plate mass — measured	H = 98.24 g , C = 97.83 g ; pair = 196.07 g
Plate pair heat capacity	172.5 J/K (196.07 g × 0.88 J/g·K, 5052 alloy)
Plate-to-plate mass difference	0.41 g (0.42%)
Test cavity	155 × 55 × 3 mm, formed by a 3 mm silicone gasket
Cavity volume	25.575 cm ³
Orientation	Heated from above
Insulation	Above and below the stack
Clamping	Identical clamps and positions for every run

Plate dimensions correspond to the BTSB-P reference geometry (280 Ah prismatic cell face), so the fixture is dimensionally representative even though it is not functionally so.

Heating from above is deliberate, for three reasons: it suppresses buoyancy-driven convection in the liquid control (Rayleigh number below the 1708 onset); a gasket failure drips molten wax away from the live heater rather than onto it; and it removes convection as a variable that differs between the liquid and solid conditions.

Note on plate mass. Both plates were weighed individually before any heat was applied, on a digital balance reading to 0.01 g. Individual masses summed to 196.07 g against a directly measured pair mass of 196.07 g — exact agreement, which establishes the balance as linear and repeatable across this range. The heat-capacity constant used in Sections 6.3 and 7.1 is computed from these measured masses, not from nominal dimensions.

The two plates differ in mass by 0.42%. Both carry the same thermocouple type in the same position and the analysis compares differences *between runs* using the same two plates throughout, so this difference is common to every run and cancels in the comparison. It is recorded rather than corrected.

4.2 Heating

Item	Specification
Heaters	3 × 12 V 40 W ceramic cartridge, 6 × 20 mm , wired in series
Supply	SEM-1250-L2, 12 V DC / 5.0 A max, centre-positive, connected directly
Voltage regulation	None. No buck converter is fitted — see below
Abort mechanism	SPST rocker switch in the positive line, within operator’s reach
Measured loop resistance	10.00 Ω (3.33 Ω per heater)
Measured delivered power	14.2 W (11.9 V × 1.19 A)
Per-heater dissipation	4.72 W against a 40 W rating — 12% of rated
Per-heater barrel heat flux	1.25 W/cm ²

The LM2596 buck converter specified in the build manual has been removed. The supply already delivers 12 V, the heaters’ design voltage, so there is nothing to step down. Fitting it would have cost approximately 1.5 V of dropout, reducing delivered power to about 10 W, and would have introduced an adjustable potentiometer whose accidental movement between runs would destroy run-to-run comparability without leaving any trace in the data. Its removal is recorded in Record G3 §2.3.

Series topology is confirmed by measurement, not only by inspection: the measured loop resistance of 10.00 Ω is a factor of nine above

the 1.11 Ω a parallel arrangement would present, and matches the series prediction to within 1%.

Two independent power figures, and which one governs

	Method	Role
Electrical	11.9 V × 1.19 A = 14.2 ± 0.5 W	Cross-check and safety verification
Thermal	Run C rise rate × plate-pair heat capacity (§6.3)	Primary — governs the energy balance

The thermal figure governs. Electrical power is the power leaving the supply; thermal power is the power that actually entered the aluminium, after losses in the wiring, the switch, the joints and to ambient from the heater barrels. The energy balance in Section 6.4 requires the latter. The electrical measurement is recorded because it independently confirms series wiring and because agreement between the two figures is itself informative about heat loss.

The electrical figure of 14.2 W was measured with heater leads at their as-supplied length of 1,010 mm. The leads are shortened before mounting, which lowers loop resistance by an estimated 0.15–0.20 Ω and raises delivered power by approximately 3%. The measurement is repeated in the trimmed configuration and the trimmed value recorded in Section 10.1, because the trimmed configuration is the one that runs the test.

4.3 Instrumentation

Item	Specification
Sensors	2 × K-type thermocouple, MAX6675 amplifier
Resolution	0.25 °C — confirmed by measurement , Record G2 §4.5
Absolute accuracy	±2.5 °C (Class 2 thermocouple tolerance)
Span verification	Both channels checked against a boiling-water reference, Record G2
Logging	Arduino Uno R3, CSV to serial at 9600 baud
Measured sampling interval	1.003 s (+0.26% of nominal), Record G1 §4.2
Fault detection	Open-thermocouple flag, verified in service during Gate 2
Thermal cut-out	None. 120 °C alarm is advisory; the operator cuts power
Electrical	In-line DC voltmeter/ammeter, uncalibrated, ±1% ±2 digits typical
TC1	Outer face, hot (upper) plate — MAX6675 #1, CS on D7
TC2	Outer face, cold (lower) plate — MAX6675 #2, CS on D8

The sampling interval is stated as a measured quantity because the analysis integrates temperature against time. A timebase error would propagate directly into every energy figure and would leave no signature in the data. At +0.26% it is two orders of magnitude below the other uncertainties here and is neglected — but it is neglected on evidence, not by assumption.

Note on accuracy. Absolute accuracy is ±2.5 °C, dominated by thermocouple tolerance. However, the primary results are **differences between runs measured by the same two sensors in the same positions**, so systematic sensor offset is common to all runs and cancels in the comparison. The differential measurement is substantially more precise than the absolute figure implies. Absolute temperatures quoted in this report should be read with the ±2.5 °C caveat; the delays should not.

Both thermocouples are on the **outer** plate faces. The calculated internal temperature gradient through 3 mm of aluminium at this heat flux is ≈0.03 °C — two orders of magnitude below sensor resolution — so outer-face placement introduces no meaningful error and avoids disturbing the test layer.

Sensor offset — how it is handled

Record G2 §4.3 originally reported a −0.63 °C difference between the channels in still air and advised applying a fixed +0.63 °C correction to every measured difference. **That advice was withdrawn.** A subsequent immersion check with both probe tips held together showed the channels agreeing to within 0.03 °C, which establishes that the −0.63 °C was a real temperature difference between two points in the room, not an instrument offset.

No fixed correction is applied to any run in this programme. Instead, the offset is re-measured at the start of every run, with both probes mounted on their plates and the fixture at ambient, and that per-run baseline is recorded in Section 10.1.

4.4 Apparatus not fitted

Item	Status	Reason
LM2596 buck converter	Not fitted	Unnecessary at 12 V; would cost ~1.5 V and add an adjustable variable — \$4.2
Heat-flux sensor	Not available	Power obtained thermally from the baseline run instead — \$6.3
Reference-standard thermometer	Not available	Span verified against a boiling-water reference only
Automatic thermal cut-out	None	Manual abort via rocker switch; alarm is advisory only

4.5 Supporting gate records

Issued before this document was dated, and before any heat was applied to the fixture.

Record	Subject	Status
G1	Software and logging chain	PASSED — no conditions
G2	Sensing verification, including boiling-water span check	PASSED
G3	Heater circuit, series topology and delivered power	PASSED, provisional pending re-measurement after lead trimming

Each gate record includes the faults found and corrected during that gate, including errors in the guidance issued to the operator. They are published unedited alongside this report.

5. METHOD

5.1 Three conditions, one fixture

Run	Layer	Phase change?	Isolates
C	Direct plate contact, thermal paste	No	Baseline
B	Liquid paraffin (mineral oil), 25.6 mL	No	Thermal resistance
A	Paraffin wax, ≈23.0 g	Yes	Resistance + latent heat

C → B

isolates thermal resistance

B → A

isolates latent heat

All three runs use the same fixture, same heater, same clamps and the same 3 mm layer thickness. Mineral oil is chosen as the control because it is chemically similar to paraffin wax (both are alkane mixtures), has thermal conductivity within approximately 30% and specific heat within approximately 10% of the wax, and remains liquid throughout the temperature range — so it has no latent heat contribution in the window of interest.

5.2 Run order and its justification

Runs are performed in the order **C, B, A**.

A trace of mineral oil remaining in the wax would add a small amount of sensible heat and slightly depress the melting point — a minor error. A trace of **wax** remaining in the **oil** would add latent heat to the control and could produce a spurious plateau in the run that must have none, causing the latent-heat contribution to be systematically **understated** and, if unnoticed, invalidating the comparison. The wax is therefore introduced last and never precedes the oil.

5.3 Controlled variables

Variable	Control
Starting temperature	All runs start within ± 2 °C of ambient and of each other
Ambient	All runs at the same time of day, within ± 2 °C
Layer thickness	Fixed by a 3 mm gasket, identical for all runs
Clamping	Same clamps, same positions, same tightness
Heater power	Same supply, no adjustable regulator fitted; voltage and current logged at the start and end of every run
Sensor position	Thermocouples not moved between runs
Sensor offset	Re-measured at ambient at the start of every run (§4.3); no fixed correction applied
Firmware	Sketch not re-uploaded during the campaign — re-upload resets the millis() timebase

5.4 Repeats

Each condition is run **twice** (six runs total) across two days, both sessions starting at the same time of day. Minimum acceptable dataset is one run per condition.

5.5 Termination

Each run terminates on the first of: TC2 reaching 85 °C; TC1 reaching 110 °C (early stop, TC2 recorded); TC1 reaching 120 °C (hard abort); 30 minutes elapsed; or any listed safety abort criterion.

6. PRE-SPECIFIED ANALYSIS

Fixed before data collection. No alternative analysis will be substituted after seeing the data.

6.1 Primary endpoint

t_{85} = time from power-on for TC2 to first reach 85 °C, by linear interpolation between the two straddling samples.

Latent-heat delay	$\Delta t_{\text{latent}} = t_{85}(\text{A}) - t_{85}(\text{B})$	← primary result
Resistance delay	$\Delta t_{\text{resistance}} = t_{85}(\text{B}) - t_{85}(\text{C})$	

6.2 Plateau detection

TC2 is smoothed with a 15-second centred moving average. The reference rise rate is the median dT/dt over the interval where smoothed TC2 lies between 40 and 55 °C. The plateau is the longest contiguous interval, within $50 \leq \text{TC2} \leq 78$ °C, over which $|dT/dt|$ is below **35%** of that reference rate. Intervals shorter than 45 seconds are rejected.

Plateau temperature is the arithmetic mean of TC2 over the detected interval.

6.3 Delivered power

Linear least-squares fit to the mean of TC1 and TC2 against time over the interval where that mean lies between 40 and 70 °C, in Run C. Delivered power **P** = **slope** × **C_plates**, where

$$C_{\text{plates}} = 196.07 \text{ g} \times 0.88 \text{ J/g}\cdot\text{K} = 172.5 \text{ J/K}$$

using the **measured** plate masses of §4.1 and the published specific heat of 5052 alloy. This constant is fixed by this document and is not adjusted after the data are seen.

This thermally derived figure — not the electrical measurement of §4.2 — is the delivered power used in the energy balance of §6.4.

6.4 Energy balance

$$\begin{aligned} E_{\text{measured}} &= P \times \text{plateau duration} \\ E_{\text{expected}} &= 23.0 \text{ g} \times 200 \text{ J/g} = 4,600 \text{ J} \\ \text{Agreement} &= E_{\text{measured}} / E_{\text{expected}} \end{aligned}$$

6.5 Handling of disagreement between repeats

If two repeats of the same condition differ by more than 15% in the primary endpoint, **both are reported separately and are not averaged**. The disagreement is reported as an uncontrolled variable.

7. PREDICTIONS

Stated before data collection.

7.1 Derivation

Cavity 155 × 55 × 3 mm	= 25.575 cm ³
× 0.90 g/cm ³ (solid paraffin)	= 23.0 g wax charge
× 190–220 J/g (paraffin latent heat)	= 4,373 – 5,064 J
Plate pair, MEASURED 196.07 g × 0.88	= 172.5 J/K
Wax charge sensible	= 48 J/K
Fixture total	= 220.5 J/K
Plateau duration = latent energy / power crossing the layer	
@ 15.0 W → 292 – 338 s (4.9 – 5.6 min)	← measured electrical power
@ 14.2 W → 308 – 357 s (5.1 – 5.9 min)	
@ 12.0 W → 364 – 422 s (6.1 – 7.0 min)	
@ 10.0 W → 437 – 506 s (7.3 – 8.4 min)	

The 14.2 W row is the electrically measured supply power and therefore an **upper bound** on the power crossing the test layer. Losses in the wiring, the switch and from the heater barrels to ambient mean the crossing power is lower, so the observed plateau is expected to fall between the 14.2 W and 10.0 W rows rather than at the top of the range. The pre-registered band of 290–510 s in §7.2 already spans this.

7.2 Predicted values

Quantity	Predicted
Delivered power (thermally derived, Run C)	10 – 15 W
Plateau duration (Run A)	290 – 510 s
Plateau temperature	58 – 68 °C
Δt _{latent} (primary endpoint)	300 – 500 s
Δt _{resistance}	100 – 250 s
t ₈₅ Run C	10 – 16 min
t ₈₅ Run B	13 – 19 min
t ₈₅ Run A	17 – 27 min
TC1 – TC2 at steady state, Run C	1 – 3 °C
TC1 – TC2 at steady state, Run B	22 – 28 °C
TC1 – TC2 at steady state, Run A	15 – 25 °C
TC1 rise in the first minute of Run C	5 – 10 °C
Intumescent onset (Section 8)	150 – 200 °C (<i>deferred — see §8</i>)

Predicted times are lower bounds. Heat loss to insulation, edges and ambient is not modelled, so measured times are expected to exceed these figures.

The first-minute TC1 rise is stated because it doubles as a live wiring check. At 14.2 W into plate H alone (98.24 g × 0.88 = **86.5 J/K**) the expected rate is **9.9 °C/min**, falling as heat reaches plate C. **A first-minute rise substantially above this band indicates a wiring fault that developed after Record G3 was issued and is grounds for abort.**

7.3 Anticipated difficulty

Mineral oil has lower thermal conductivity than wax (≈0.14 vs ≈0.25 W/m·K). At the measured 14.2 W the predicted gradient across the layer in Run B reaches ≈25 °C, placing TC1 near 110 °C when TC2 reaches 85 °C — at the early-stop limit and within 10 °C of the hard abort. **Run B is therefore the run most likely to terminate early.** A truncated Run B remains usable provided the TC2 value reached is recorded.

8. SECOND TEST — INTUMESCENT ONSET

STATUS AT PRE-REGISTRATION: DEFERRED — MATERIAL NOT OBTAINED.

Expandable graphite has not been sourced as of this issue. The supplier approached (Wedge INTUF14) did not respond and no alternative has been secured within the programme window.

This section and hypothesis H4 are pre-registered but not under test in the current campaign. They are retained, unaltered, so that if material is obtained later the test can be run against criteria fixed before the material was in hand. If the test is not performed, Section 10.3 will record it as not performed, with this reason. **It will not be quietly removed.**

Expandable graphite (sulphuric-acid intercalated) is heated in a steel vessel on a hotplate under a slow ramp. TC1 is attached to the vessel base; TC2 is buried in the flake bed.

Primary endpoint: TC2 at first visible expansion, declared aloud on video at the moment of observation. n = 3.

Secondary: free volumetric expansion ratio, by loose uncompacted poured volume before and after.

The free expansion ratio is not the product figure. Free expansion of unconstrained flakes is not comparable to constrained expansion of a 0.40 mm layer against back-pressure in a clamped gap. It is reported as free expansion, with that distinction stated.

9. FALSIFICATION CRITERIA

What result would show the hypothesis to be wrong. Stated before data collection.

#	Hypothesis	Falsified if
H1	A phase-change layer delays heat transfer relative to a non-phase-change layer of identical thickness	$\Delta t_{\text{latent}} \leq 60 \text{ s}$ across at least two repeats
H2	The delay is attributable to latent heat, not to a difference in thermal resistance	No plateau detected, or energy balance agreement outside 50-150%
H3	Absorption occurs at approximately constant temperature in the expected band	No plateau detected, or plateau temperature outside 55-72 °C
H4 (deferred)	The intumescent activates at a temperature that does not overlap the phase-change band	Onset below 100 °C , or the two measured bands overlap

H1 to H3 are under test in this campaign. H4 is deferred — the material has not been obtained (§8). H4’s criteria are fixed here and will not be revised if and when the material becomes available.

9.1 Results that would NOT falsify

Stated to prevent a false negative being read as failure:

- A plateau **shorter** than predicted, with the energy balance closing, indicates a smaller effective charge — not an absent mechanism.
- Times **longer** than predicted are expected, since heat loss is not modelled.
- Run B terminating early on the TC1 limit is a predicted apparatus limitation (§7.3), not a result.
- Disagreement with a supplier datasheet is a finding about the material, not about the method.

9.2 Commitment

If any falsification criterion is met, that outcome will be reported in Revision B of this document and in all derived material, with no change to the criteria stated above.

10. RESULTS

To be completed after testing. Blank at pre-registration.

10.1 Runs performed

Fixture constants as measured

	Value	When measured
Plate H mass	98.24 g	before first run
Plate C mass	97.83 g	before first run
Plate pair, weighed together	196.07 g	before first run
Plate pair heat capacity, C_plates	172.5 J/K	computed
Supply V / A, untrimmed heater leads	11.9 V / 1.19 A → 14.2 W	Record G3
Supply V / A, trimmed heater leads	____ V / ____ A	before first run
Delivered electrical power, trimmed	____ W	before first run

Runs

Run	Date	Ambient °C	Start TC2 °C	Offset TC1–TC2 at start	V / A start	V / A end	Stop reason	t ₈₅	CSV
C-1									
B-1									
A-1									
C-2									
B-2									
A-2									

10.2 Primary endpoint

	Value
Δt _{latent} (A – B)	_____ s
Δt _{resistance} (B – C)	_____ s
Plateau duration	_____ s
Plateau temperature	_____ ± _____ °C
Delivered power, thermally derived (governs)	_____ W
Delivered power, electrically measured (cross-check)	_____ W
Ratio thermal / electrical	_____ %
E _{measured}	_____ J
E _{expected}	4,600 J
Agreement	_____ %

The ratio of thermal to electrical power is reported because it quantifies total system heat loss. A ratio near 100% would indicate a well-insulated fixture; a low ratio quantifies how much power never reached the aluminium. Either result is reported as found.

10.3 Intumescent

Status: _____ (performed / not performed — if not performed, state why)

	Value
Onset (TC2, first visible expansion)	_____ ± _____ °C, n = ____
Vessel-base lag (TC1 – TC2) at onset	_____ °C
Free expansion ratio	_____ x

10.4 Aborted, failed or excluded runs

All runs attempted are listed here, including those excluded, with the reason.

10.5 Figures

- **Figure 1** — TC2 against time, Runs C / B / A, plateau shaded, 85 °C marked
- **Figure 2** — Measured temperature bands: phase change and intumescent onset

11. ANALYSIS

To be completed after testing.

12. LIMITATIONS

Applicable regardless of outcome.

1. Proxy PCM (paraffin wax), not the specified RT64HC.

2. Bench charge $\approx 3.3\times$ the latent energy of a production pad.
3. No cells; heat supplied resistively at approximately constant power, whereas thermal runaway is a transient with a far higher peak flux.
4. Two-point temperature measurement; no heat-flux sensor.
5. Delivered power obtained two ways — thermally from the baseline run (governing) and electrically at the supply (cross-check). Neither is traceable to a calibrated standard. The electrical meter is an uncalibrated commercial module of $\pm 1\%$ ± 2 digits typical accuracy.
6. Absolute accuracy ± 2.5 °C; differential measurement is more precise but not quantified by calibration against a reference standard. Span verified only against a boiling-water reference, i.e. at a single elevated point.
7. Uncontrolled heat loss to insulation, edges and ambient. Quantified only indirectly, by the thermal-to-electrical power ratio in §10.2.
8. Single fixture, single operator, not independently witnessed, not accredited.
9. **No automatic thermal cut-out.** The 120 °C limit is a printed alarm; the interlock is the operator's hand on a manual switch.
10. **The intumescent test (§8, H4) is deferred — material not obtained.** Four of BTSB's five layers remain entirely untested by this programme; this campaign addresses one.
11. Free intumescent expansion, if later measured, is by uncompacted poured volume, $\pm 20\%$.
12. Constrained intumescent expansion in the product geometry not measured.

13. CONCLUSIONS

To be completed after testing.

14. DATA AVAILABILITY

All raw CSV files, unedited, together with run video, the completed field log sheets, and gate records G1, G2 and G3, are published at:

Failed and aborted runs are included. Gate records are published including the fault sections, unedited.

15. SIGNATURE

Prepared by	S. Mohammed Idreesh
Position	Founder, CIP — Compute Infrastructure Platform
Signature	_____
Date of pre-registration (Rev A)	25 August 2026
Second person present	_____
Date of final issue (Rev B)	_____

HOW TO TIMESTAMP THIS DOCUMENT

Pre-registration only means something if the date can be shown to precede the data. Before your first run, do at least one of these — all are free and take minutes:

1. **Email the PDF to yourself and to one other person.** The mail server's timestamp is independent of you.
2. **Upload to Google Drive** and share a public read-only link. Drive records the creation date.
3. **Commit it to a public GitHub repository.** The commit timestamp is cryptographically chained and is the strongest of the three.

Then put that link in Section 14 of the final revision, so a reader can verify for themselves that the predictions preceded the results.

Report CIP-BTSB-BT-001 Rev A · Pre-registration · Contains no results. Governing values per ECN-001. All build-manual performance figures remain [GEN-0 VALIDATION REQUIRED].